Our house is burning down and we're blind to it

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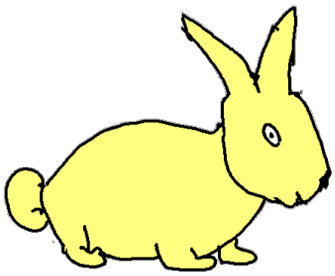
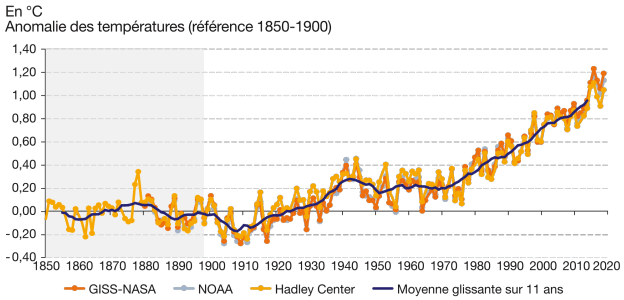
APM_4MA04_TP – ODE





1 Motivation

My rabbit Bibi complains of being too hot. I told him he'd better go to the groomer to get his fleece cut but he replied that the problem is more fundamental: according to him, the climate is getting warmer!

If we listen to the IPCC, this would be due to the carbon cycle going haywire. On the website `https://exchange.iseesystems.com/public/globeprogam/global-carbon-cycle-model-with-feedbacks/index.html` you can find a carbon cycle simulator. The equations seem correct but nothing says that the resolution algorithm used in this simulator is correct. In fact, I'm quite skeptical about the results and I'm afraid that environmental hackers may have tampered with the results. I would prefer a program in which we control all the elements. It should indeed be ensured that what is calculated is in good agreement with reality. It seems that the concepts of consistency and stability are the key to guarantee the reliability of numerical methods...

I would need your help to confirm or refute the conclusions of the IPCC and to simulate the temperature variation for the years to come. Then we could try and see how we can modify our carbon dioxyde emissions profile for Bibbi to stop complaining.

2 Learning Objectives

- Implement numerical methods for solving differential equations.
- Formalize a Cauchy problem and study the properties of the functions it involves.
- Demonstrate the convergence of a numerical method from a theoretical perspective and empirically.
- Compare the performance of numerical methods.

3 Resources

- Handout
- Course Videos
- Worksheets
- File `carbone.py`
- Website of the carbon cycle simulator
- Questions to the teachers

4 Deliverables

For Wednesday April 8th, 2026 at 11:59 p.m., you will upload to moodle:

- a presentation of 5 slides that describe your method and justify that it works correctly,
- the code of your numerical method in the language of your choice.

On Friday April 10th, you are going to present your work in front of the class.

5 Working plan

Step 1 - 5mn - Look at the document, organize the group

Step 2 - 10mn - Read in detail and look at the equations
Common reading:
What questions does the subject raise?
In your opinion what are the important words of the subject?

Step 3 - 5mn - Understand and reformulate the problem
Summarize the identified problem (including deliverables) in written form
Appoint a rapporteur who will describe your summary.

Step 4 - 10mn- General sharing between all groups.

Step 5 - 15mn- Possible solutions
Generate possible ideas/actions to take to address the problem.
First individually for 5 minutes then shared within the group.

Step 6 Establish the necessary learning and the sub-problems to be addressed to answer the question

Step 7 Establish an action plan
Distribute tasks taking into account time constraints.
Put this distribution in writing.

Step 8 Personal work

Step 9 Submit the deliverables

Step 10 Present your work in front of the class